

Model Lab

Every prior, same slate — 2026-09-10

SERVED PRIOR

Core (KNN+SVR)

PRIORS COMPARED

10

HELD-OUT PA

35,068

Held-Out Scores

Fitted on plate appearances before 2026-08-06, scored on the 35,068 that came after (actual HR rate 2.96%). No outcome in the test window was visible during fitting.

Metrics are per plate appearance; the probabilities lower down are per game.

Read AUC first — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5673	1.39×	0.02866	0.13256	0.00369
KNN comps Prior is the pooled rate of the nearest comparables.	0.5870	1.57×	0.02865	0.13267	0.00524
SVR Support-vector regression on the same feature matrix.	0.5791	1.43×	0.02873	0.13389	0.00863
Random forest Forest regression; a learned similarity metric.	0.5813	1.45×	0.02870	0.13337	0.00731
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5865	1.55×	0.02865	0.13276	0.00553
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5860	1.50×	0.02865	0.13264	0.00587
Linear regression Ordinary least squares on the same features.	0.5856	1.46×	0.02867	0.13299	0.00657
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5867	1.48×	0.02868	0.13275	0.00697
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5855	1.44×	0.02867	0.13275	0.00714
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5803	1.45×	0.02876	0.13406	0.00912

Blend weights (forest variant): KNN 0.75, SVR 0.00, RF 0.25

Shrinkage K per prior: league 248, knn 248, svr 248, rf 248, core 248, ensemble 248, linear 248, logistic 248, nn 248, xgb 248 — estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-Sample Subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,602 PA, actual HR rate 2.50%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5452	1.45×	0.02438	0.11722	0.01065
KNN comps Prior is the pooled rate of the nearest comparables.	0.5676	1.33×	0.02447	0.12293	0.01432
SVR Support-vector regression on the same feature matrix.	0.5639	1.22×	0.02494	0.12851	0.02397
Random forest Forest regression; a learned similarity metric.	0.5670	1.00×	0.02470	0.12514	0.01994
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5652	1.33×	0.02452	0.12340	0.01477
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5663	1.33×	0.02451	0.12205	0.01419
Linear regression Ordinary least squares on the same features.	0.5815	1.11×	0.02466	0.12473	0.01829
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5737	1.22×	0.02467	0.12234	0.01880
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5763	1.00×	0.02458	0.12173	0.01708
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5657	1.11×	0.02525	0.13026	0.03035

Slate Agreement

Flat league · 4/6 shared 0/6 hit

Kyle Schwarber
Junior Caminero
Jeremy Peña
Yordan Alvarez
Ronald Acuña Jr.
Bryce Harper

SVR · 5/6 shared 0/4 hit

Kyle Schwarber
Junior Caminero
Yordan Alvarez
Corey Seager
Aaron Judge
Matt Olson

Core (KNN+SVR) · 6/6 shared 0/4 hit

Kyle Schwarber
Yordan Alvarez
Corey Seager
Junior Caminero
Aaron Judge
Ronald Acuña Jr.

Linear regression · 5/6 shared 0/3 hit

Kyle Schwarber
Yordan Alvarez
Lazaro Montes
Corey Seager
Junior Caminero
Aaron Judge

Neural net (torch) · 5/6 shared 0/4 hit

KNN comps · 5/6 shared 0/5 hit

24.7%	Kyle Schwarber	30.6%
21.3%	Yordan Alvarez	24.7%
20.8%	Corey Seager	24.2%
19.5%	Junior Caminero	23.2%
18.5%	Ronald Acuña Jr.	21.7%
18.5%	Jeremy Peña	21.4%

Random forest · 5/6 shared 0/3 hit

34.0%	Kyle Schwarber	31.5%
28.0%	Yordan Alvarez	27.0%
27.3%	Corey Seager	24.4%
23.6%	Lazaro Montes	24.0%
23.6%	Junior Caminero	23.8%
23.1%	Aaron Judge	23.7%

Forest blend · 5/6 shared 0/4 hit

31.1%	Kyle Schwarber	30.8%
25.1%	Yordan Alvarez	25.3%
24.1%	Corey Seager	24.3%
23.9%	Junior Caminero	23.3%
21.6%	Aaron Judge	21.9%
21.5%	Jeremy Peña	21.4%

Logistic regression · 5/6 shared 0/3 hit

32.3%	Kyle Schwarber	37.3%
25.8%	Yordan Alvarez	29.2%
25.4%	Lazaro Montes	28.3%
23.7%	Aaron Judge	25.8%
23.6%	Junior Caminero	25.0%
23.5%	Corey Seager	24.2%

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)	XGBoost
1. Kyle Schwarber slate Philadelphia Phillies · vs RHP · 44 HR / 628 PA	24.7%	30.6%	34.0%	31.5%	31.1%	30.8%	32.3%	37.3%	32.7%	32.9%
2. Yordan Alvarez Houston Astros · vs RHP · 38 HR / 632 PA	19.5%	24.7%	27.3%	27.0%	25.1%	25.3%	25.8%	29.2%	27.5%	27.7%
3. Corey Seager slate Texas Rangers · vs RHP · 17 HR / 366 PA	17.3%	24.2%	23.6%	24.4%	24.1%	24.3%	23.7%	24.2%	24.6%	24.8%
4. Junior Caminero slate Tampa Bay Rays · vs RHP · 39 HR / 635 PA	21.3%	23.2%	28.0%	23.8%	23.9%	23.3%	23.6%	25.0%	24.9%	25.1%
5. Aaron Judge slate New York Yankees · vs LHP · 17 HR / 268 PA	13.2%	21.3%	23.6%	23.7%	21.6%	21.9%	23.5%	25.8%	24.2%	24.4%
6. Ronald Acuña Jr. slate Atlanta Braves · vs LHP · 16 HR / 408 PA	18.5%	21.7%	20.8%	20.7%	21.5%	21.4%	21.1%	21.5%	21.5%	21.7%
7. Jeremy Peña slate Houston Astros · vs RHP · 17 HR / 416 PA	20.8%	21.4%	21.5%	21.5%	21.4%	21.4%	20.7%	20.5%	20.6%	20.8%

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)	XGBoost
8. Matt Olson Atlanta Braves · vs LHP · 37 HR / 632 PA	16.9%	21.0%	23.1%	22.2%	21.3%	21.3%	21.2%	23.7%	22.8%	22.5%
9. Bryce Harper Philadelphia Phillies · vs RHP · 27 HR / 623 PA	18.5%	21.2%	21.5%	21.8%	21.3%	21.4%	21.2%	21.0%	21.4%	21.3%
10. Cal Raleigh Seattle Mariners · vs RHP · 21 HR / 462 PA	16.1%	21.3%	19.0%	19.6%	20.9%	20.9%	22.0%	23.9%	21.4%	21.2%
11. Miguel Vargas Chicago White Sox · vs LHP · 30 HR / 630 PA	14.3%	20.0%	19.2%	18.8%	19.9%	19.7%	19.4%	20.7%	19.9%	19.6%
12. Joc Pederson Texas Rangers · vs RHP · 25 HR / 421 PA	16.1%	19.3%	19.5%	19.7%	19.4%	19.4%	18.8%	18.9%	19.1%	18.9%
13. Ben Rice New York Yankees · vs LHP · 36 HR / 599 PA	14.4%	18.9%	20.9%	19.7%	19.2%	19.1%	18.5%	19.3%	19.2%	19.0%
14. Oneil Cruz Pittsburgh Pirates · vs RHP · 17 HR / 363 PA	13.9%	19.1%	19.4%	16.9%	19.1%	18.5%	17.2%	16.8%	17.9%	17.6%
15. Hunter Goodman Colorado Rockies · vs RHP · 37 HR / 517 PA	16.3%	18.9%	20.0%	19.5%	19.0%	19.0%	18.7%	20.0%	19.9%	19.7%
16. Lazaro Montes Seattle Mariners · vs RHP · 2 HR / 20 PA	11.8%	19.5%	16.0%	24.0%	19.0%	20.6%	25.4%	28.3%	22.1%	21.8%
17. Mickey Moniak Colorado Rockies · vs RHP · 22 HR / 397 PA	14.6%	18.4%	20.3%	18.2%	18.7%	18.4%	18.0%	19.5%	19.1%	18.9%
18. Cam Smith Houston Astros · vs RHP · 19 HR / 509 PA	16.4%	18.7%	18.6%	18.2%	18.6%	18.5%	18.3%	18.5%	18.9%	18.7%
19. Rafael Flores Jr. Pittsburgh Pirates · vs RHP · 8 HR / 114 PA	12.3%	18.0%	21.8%	18.5%	18.5%	18.1%	19.3%	18.7%	19.8%	19.5%
20. Ryan Vilade Tampa Bay Rays · vs RHP · 11 HR / 236 PA	16.4%	18.4%	18.6%	18.8%	18.4%	18.5%	17.6%	17.1%	18.7%	18.5%
21. Austin Riley Atlanta Braves · vs LHP · 16 HR / 570 PA	13.6%	18.1%	16.5%	15.8%	17.9%	17.5%	15.9%	16.9%	16.9%	16.7%
22. Lane Thomas	15.5%	17.8%	16.4%	15.8%	17.6%	17.3%	15.4%	15.4%	15.4%	15.4%

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)	X
37. Yandy Díaz Tampa Bay Rays · vs RHP · 22 HR / 625 PA	15.8%	14.8%	14.9%	15.3%	14.8%	14.9%	15.5%	15.3%	14.7%	
38. Wyatt Langford Texas Rangers · vs RHP · 16 HR / 405 PA	14.3%	14.9%	14.0%	14.2%	14.8%	14.8%	13.5%	13.6%	14.1%	
39. Michael Harris II Atlanta Braves · vs LHP · 24 HR / 557 PA	14.1%	14.0%	16.0%	13.6%	14.3%	13.9%	14.5%	16.1%	15.1%	
40. Jose Altuve Houston Astros · vs RHP · 15 HR / 505 PA	15.8%	14.1%	15.3%	14.4%	14.3%	14.1%	13.4%	13.7%	13.5%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context.